CASE-STUDY PRACTICE | CAPSTONE 2 | MOCK INTERVIEW EDITION

Mock Interview Edition

Ad CTR Prediction: Online Learning at Scale

18 scenarios with their follow-up probes and no answers. Use it to rehearse aloud, with a partner, or against a timer, then check yourself against the Study Edition.

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HOW THIS EDITION IS ORGANIZED

Section A, round plans: 5 ready-made sessions of five scenarios each, by theme. **Section B, question sheets:** every scenario followed by all of its probes, in order, each with its own notes box. The Study Edition page that holds the answers is printed under each scenario. **Section C, score sheet:** one row per scenario to track attempts. Question numbers and part numbers are identical in the Study Edition.

THREE WAYS TO USE IT

1. With a partner. The partner reads the scenario aloud, then asks the probes one at a time, moving on only after you have answered. **2. Alone, out loud.** Cover everything below the current probe with a sheet of paper, answer aloud for about two minutes, then slide the paper down to the next probe. **3. Alone, in writing.** Jot your key points in the notes box under each probe before looking at the next one, then compare with the Study Edition page printed under the scenario.

HOW TO GRADE A PROBE

Solid: you stated the main point, gave a number or a concrete example where one was possible, and said how you would check it. **Partial:** right direction but vague, or a number without the reasoning. **Missed:** you did not reach it, or reached the wrong conclusion. The last probe of every scenario asks for the general lesson in one sentence. Suggested pacing is about 12 minutes per scenario. Reading the answer only after you have committed to your own is what makes the practice work.

A NOTE ON THE PACING, GRADING AND ROUNDS

The 12-minute pacing, the three-level grading and the round compositions are the author's suggestions for practice, not an industry standard. The scenarios are written for this document and are not taken from a real company question bank. Where a scenario quotes a figure, the Study Edition says where it comes from.

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SECTION A

Round plans

Each round has five scenarios at about 12 minutes each, a 60-minute session. Mix and match freely. Click any scenario to jump to its question sheet; the number on the right is the page.

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02 When does a click count? Labels 5
03 Size the prediction traffic Estimation 7
04 Does 16-bit storage save 75%? Sizing 8
16 Calibrate, and say how sure the model is Evaluation 25

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09 Why not plain gradient descent with an L1 penalty? Algorithm choice 15
14 Replicate a model that never stops changing Serving 22
17 Ten times the traffic Scale 27

SECTION B

Question sheets

Each scenario is followed by its probes in order. Indentation shows how deep the follow-up goes.

PART 1

Framing the problem: probabilities, labels and the sizing

Why the auction needs a calibrated probability, how late clicks bend the labels, and what a prediction costs in compute and in memory.

01 Why a probability, and not a yes or no?

FRAMING

SCENARIO

A teammate proposes treating ad click prediction as a classifier that outputs click or no click, judged by accuracy. The paper says that an auction takes the advertisers' bids and an estimate of P(click | q, a) as inputs. Argue for or against the teammate's design.

OPENING PROBE

What does the auction need from the model?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

Why is a yes or no not enough?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What does accuracy report for a model that knows nothing?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What does the paper use instead?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 7

02 When does a click count?

LABELS

SCENARIO

Impressions and clicks come from two streams. The paper says its data is provided by a streaming service based on Photon and says nothing about how long it waits for a click. You must choose a join window. A colleague says: wait a long time, so no label is ever wrong.

OPENING PROBE

What does the paper actually say about the join?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What are the two errors?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

How big can the bias be?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

How would you choose the window?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 8

03 Size the prediction traffic

ESTIMATION

SCENARIO

The paper says only that a typical model predicts on billions of events a day and that its own system predicts many billions of times a day. Plan for 10 billion predictions a day with 300 non-zero features each. Is the pressure on compute or on memory?

OPENING PROBE

What are the rates?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

How many queries is that?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

How much compute is a prediction?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

Then what is large?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 9

04 Does 16-bit storage save 75%?

SIZING

SCENARIO

The paper says that storing coefficients as 16-bit fixed point instead of 64-bit floats saves 75% of the RAM for coefficient storage with no measurable loss. A manager reads that and plans the serving fleet's memory as one quarter of the float size. The model has 2 billion non-zero coefficients.

OPENING PROBE

What does the paper's figure cover?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What else is in the served table?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What is the whole-model saving?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What does that do to the fleet?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 10

PART 2

Features and data: hashing, rare features and subsampling

Choosing how to identify billions of features, deciding which to track, and shrinking the training stream without biasing it.

05 Hash the features, or keep them exact?

DESIGN CHOICE

SCENARIO

An engineer wants to cut memory by hashing every feature into a fixed number of buckets, as papers by Weinberger et al. and Chapelle did with 2 to the power of 24 buckets. The CTR paper tried it. Decide, using the paper and your own arithmetic.

OPENING PROBE

What does the paper report?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What is the difference between the two layouts?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

Which wins here?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What else counts?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 11

06 Which rare features deserve a slot?

MEMORY

SCENARIO

In some of the paper's models, half the unique features occur only once in billions of examples. You cannot tell in advance which features will be rare, and a pre-pass that drops rare features costs an extra read and write of the data. Pick a method.

OPENING PROBE

Why not pre-filter the data?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What are the two online methods?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What does the paper's table say?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

Check the numbers against the half-are-singletons fact.

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 12

07 Subsample the non-clicked queries

DATA

SCENARIO

Training on every impression is expensive and clicks are rare. The paper keeps every query with at least one click and a fraction r of the rest, and weights the kept unclicked queries by 1 over r. You have 100 million queries a day with 3 ads each and a 2% click rate per ad.

OPENING PROBE

How many queries have a click?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What is kept at r = 0.10?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

Do the weights restore the truth?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What goes wrong without weights?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 13

08 Review the sampler

CODE REVIEW

SCENARIO

A colleague writes this sampler for the training stream and says it implements the paper's rule. The stream has one event per ad shown, with 3 ads per query and a 2% click rate per ad. The loss curve looks fine and the pipeline is fast. In an A/B test the model's predicted CTRs are far above the observed ones.

```
import random

def subsample(events, r=0.1):
    # events: one record per ad shown; fields query_id, clicked, features
    for e in events:
        if e.clicked or random.random() < r:
            yield e, 1.0          # (example, importance weight)
```

OPENING PROBE

What does the code do?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What is the first fault?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What is the second fault?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

State the fix.

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 14

PART 3

The learning algorithm: FTRL-Proximal and per-coordinate rates

Why plain gradient descent is not enough, how to read the paper's comparison table, and what one update does to one coordinate.

09 Why not plain gradient descent with an L1 penalty?

ALGORITHM CHOICE

SCENARIO

Serving needs a small model, so the team wants sparsity. They propose OGD with an L1 penalty. The paper's Table 1 compares FTRL-Proximal with RDA, FOBOS and OGD-Count. Decide, and say how much you trust the table.

OPENING PROBE

Why does sparsity matter here?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What goes wrong with OGD and L1?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What does FTRL-Proximal change?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

How do you read Table 1?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 16

10 Why does each coordinate get its own learning rate?

ALGORITHM

SCENARIO

The paper reports that per-coordinate learning rates cut AucLoss by 11.2% against a single global rate. A skeptic says that this must be a tuning artifact: both models just need a better learning rate. Explain the mechanism and answer the skeptic.

OPENING PROBE

What is wrong with one global rate?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 5

What do the numbers look like?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 5

What is the paper's rate?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 5

Is the 11.2% a tuning artifact?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 5

What does 11.2% mean in absolute terms?

YOUR NOTES: key points you would make

FOLLOW-UP 5 OF 5

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 17

11 Trace one coordinate by hand

CALCULATION

SCENARIO

An interviewer gives you FTRL-Proximal with alpha = 0.1, beta = 1, L1 = 0.5 and L2 = 0. One feature has value 1. Its weight starts at 0, so the predicted probability is 0.5. The impression is clicked, twice in a row. Compute z, n and the weight after each click, and say what it shows.

OPENING PROBE

What happens at the first click?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What happens at the second click?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What does that show about rare features?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What would change with no L1 penalty?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 18

12 Review the FTRL update

CODE REVIEW

SCENARIO

A colleague implements FTRL-Proximal. On a test with tiny data it produces the right signs, the sparsity looks right and the loss goes down. In a larger comparison against the reference implementation, the weights are always slightly different. Find out why.

```
def update(self, x, y):
    # self._weight(i) reads self.z[i] and self.n[i]
    p = self.predict(x)
    for i, v in x.items():
        g = (p - y) * v
        old_n = self.n.get(i, 0.0)
        self.n[i] = old_n + g * g
        sigma = (math.sqrt(self.n[i]) - math.sqrt(old_n)) / self.alpha
        self.z[i] = self.z.get(i, 0.0) + g - sigma * self._weight(i)
```

OPENING PROBE

Do the first two steps agree with the hand trace?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 5

Where do they begin to differ?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 5

What is the fault?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 5

Which test would have caught it?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 5

State the fix.

YOUR NOTES: key points you would make

FOLLOW-UP 5 OF 5

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 19

13 Counts instead of squared gradients, and many variants at once

MEMORY

SCENARIO

Per-coordinate rates need the sum of squared gradients for each feature, a second number to store. The paper replaces it with two counts, and also trains many model variants together. Explain what is approximated, and where the memory really goes down.

OPENING PROBE

What is the approximation?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 5

Check it on numbers.

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 5

Does it save memory by itself?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 5

What does sharing across variants save?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 5

What about the single value structure?

YOUR NOTES: key points you would make

FOLLOW-UP 5 OF 5

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 20

PART 4

Serving, evaluation and operations

Replicating a continuously changing model, reading metrics with their noise, calibrating, scaling traffic and guarding the learner.

14 Replicate a model that never stops changing

SERVING

SCENARIO

Training updates the weights after every example. The model must be copied to many data centers. The paper says only that trained models are replicated to many data centers for serving. Design how the copies are kept fresh, and what is stored where.

OPENING PROBE

What is in the paper, and what is not?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 5

What does a serving replica hold?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 5

What does a snapshot cost?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 5

What does a delta cost?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 5

What if a replica loses its link to training?

YOUR NOTES: key points you would make

FOLLOW-UP 5 OF 5

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 22

15 How good is a metric, and how noisy?

EVALUATION

SCENARIO

The paper evaluates with progressive validation, aggregated hourly, and also by slice (country, query topic, layout). A teammate says a 1% relative improvement in a slice is a clear win. Explain what progressive validation measures and how much noise a slice carries.

OPENING PROBE

What is progressive validation?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 5

What are its limits?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 5

How noisy is the whole stream?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 5

How noisy is a small slice?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 5

Why does the paper add a visualization?

YOUR NOTES: key points you would make

FOLLOW-UP 5 OF 5

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 23

16 Calibrate, and say how sure the model is

EVALUATION

SCENARIO

The auction needs honest probabilities, and it also benefits from knowing how sure the model is about each one. The paper adds a calibration layer and proposes an uncertainty score. Explain both, and what limits them.

OPENING PROBE

What does calibrated mean?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 5

What are the two fixes?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 5

How does isotonic regression treat noisy buckets?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 5

What cannot calibration promise?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 5

What does the uncertainty score do?

YOUR NOTES: key points you would make

FOLLOW-UP 5 OF 5

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 24

17 Ten times the traffic

SCALE

SCENARIO

Imagine 100 billion predictions a day. A manager asks what breaks first, and whether to add machines or change the design.

OPENING PROBE

What happens to rates?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 5

What happens to the training stream?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 5

What happens to the model?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 5

What happens to replication?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 5

What happens to the measurements?

YOUR NOTES: key points you would make

FOLLOW-UP 5 OF 5

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 25

18 Fake clicks and a broken signal

RISK

SCENARIO

An online learner trusts the clicks it sees. A security reviewer asks what happens if a ring generates fake clicks, and an engineer asks what happens if an input signal silently changes meaning. The paper discusses neither fraud nor attacks.

OPENING PROBE

How far can fake clicks move a rate?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

Where would you defend?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What does a silently changed signal do?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

How would you notice?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 26

SECTION C

Score sheet

Write how many probes were solid out of the total, per attempt. Revisit anything under half after reading the Study Edition's concepts box.

No.	Scenario	Probes	Attempt 1	Attempt 2	Attempt 3
01	Why a probability, and not a yes or no?	5			
02	When does a click count?	5			
03	Size the prediction traffic	5			
04	Does 16-bit storage save 75%?	5			
05	Hash the features, or keep them exact?	5			
06	Which rare features deserve a slot?	5			
07	Subsample the non-clicked queries	5			
08	Review the sampler	5			
09	Why not plain gradient descent with an L1 penalty?	5			
10	Why does each coordinate get its own learning rate?	6			
11	Trace one coordinate by hand	5			
12	Review the FTRL update	6			
13	Counts instead of squared gradients, and many vari...	6			
14	Replicate a model that never stops changing	6			
15	How good is a metric, and how noisy?	6			
16	Calibrate, and say how sure the model is	6			
17	Ten times the traffic	6			
18	Fake clicks and a broken signal	5			